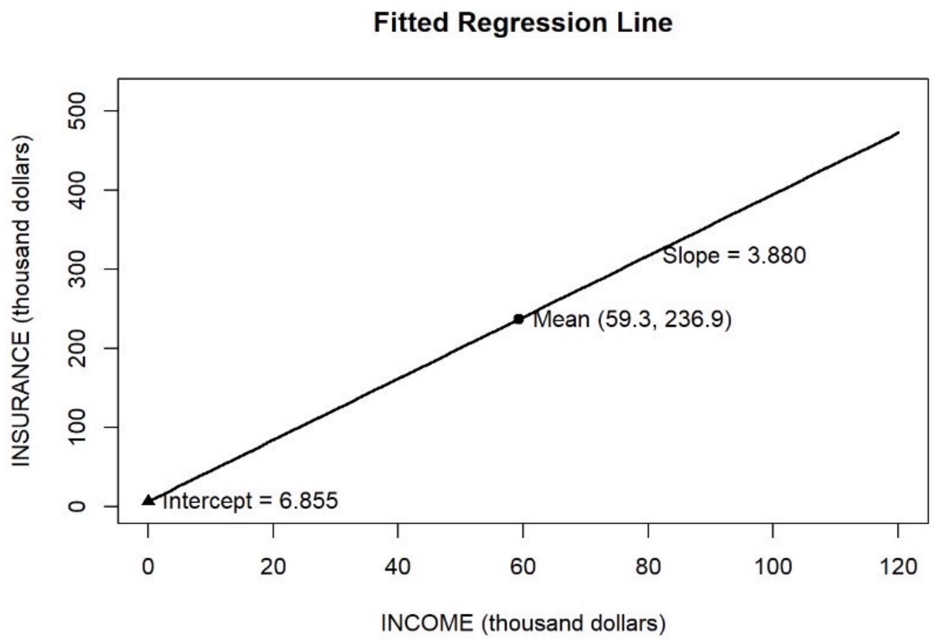
3.18 A life insurance company examines the relationship between the amount of life insurance held by a household and household income. Let *INCOME* be household income (thousands of dollars) and *INSURANCE* the amount of life insurance held (thousands of dollars). Using a random sample of $N = 20$ households, the least squares estimated relationship is

$$\widehat{INSURANCE} = \underset{(7.383)}{6.855} + \underset{(0.112)}{3.880}INCOME \qquad \text{(se)}$$

a. Draw a sketch of the fitted relationship identifying the estimated slope and intercept. The sample mean of *INCOME* = 59.3. What is the sample mean of the amount of insurance held? Locate the point of the means in your sketch.

$\bar{X} = 59.3$, $\bar{Y} = 6.855 + 3.880 \times 59.3 = 236.939$



b. How much do we estimate that the average amount of insurance held changes with each additional \$1000 of household income? Provide both a point estimate and a 95% interval estimate. Explain the interval estimate to a group of stockholders in the insurance company.

$\hat{\beta}_2 \pm t_{0.975,18} \cdot SE$, $df = 20 - 2 = 18$

$= 3.880 \pm 2.102 \times 0.112$

$= 3.880 \pm 0.235$

$\Rightarrow$ 95% C.I. $= [3.645, 4.115]$

In repeated sampling, 95% of intervals would be expected to capture the true population parameter.
With 95% confidence, an increase of \$1000 in household income is associated with an increase of between \$3.645 and \$4.115 in life insurance holdings on average.

c. Construct a 99% interval estimate of the expected amount of insurance held by a household with \$100,000 income. The estimated covariance between the intercept and slope coefficient is −0.746.

$Cov(\hat{\beta}_1, \hat{\beta}_2) = -0.746$

$\hat{Y}_0 = 6.855 + 3.880 \times 100 = 394.855$

$Var(\hat{Y}_0) = Var(\hat{\beta}_1) + x_0^2 \, Var(\hat{\beta}_2) + 2x_0 \, Cov(\hat{\beta}_1, \hat{\beta}_2)$

$= (7.383)^2 + 100^2 (0.112)^2 + 2 \cdot 100 \cdot (-0.746)$

$= 30.7487$

$SE = \sqrt{30.7487} = 5.5452$

99% C.I.: $t_{0.005, 18} \approx 2.878$

$394.855 \pm 2.878 \times 5.5452$

$= 394.855 \pm 15.959$

$\Rightarrow [378.896, 410.814]$ #

d. One member of the management board claims that for every \$1000 increase in income the average amount of life insurance held will increase by \$5000. Let the algebraic model be $INSURANCE = \beta_1 + \beta_2 INCOME + e$. Test the hypothesis that the statement is true against the alternative that it is not true. State the conjecture in terms of a null and alternative hypothesis about the model parameters. Use the 5% level of significance. Do the data support the claim or not? Clearly, indicate the test statistic used and the rejection region.

$t_{0.975, 18} = 2.101$

$H_0: \beta_2 = 5$

$H_1: \beta_2 \neq 5$, $\quad t = \frac{3.880 - 5}{0.112} = -10$, $\quad RR = \{|t| \geq 2.101\}$

$\because$ $|t| = 10 > 2.101$

$\therefore$ reject H_0

$\Rightarrow$ There is sufficient evidence to suggest $\beta_2 \neq 5$ at 5% significance level

$\Rightarrow$ We deny the member's claim.

e. Test the hypothesis that as income increases the amount of life insurance held increases by the same amount. That is, test the null hypothesis that the slope is one. Use as the alternative that the slope is larger than one. State the null and alternative hypotheses in terms of the model parameters. Carry out the test at the 1% level of significance. Clearly indicate the test statistic used, and the rejection region. What is your conclusion?

$H_0: \beta_2 = 1$

$H_1: \beta_2 > 1$, $\quad t = \frac{3.880 - 1}{0.112} = 25.71$, $\quad RR = \{t > t_{0.01, 18} = 2.552\}$

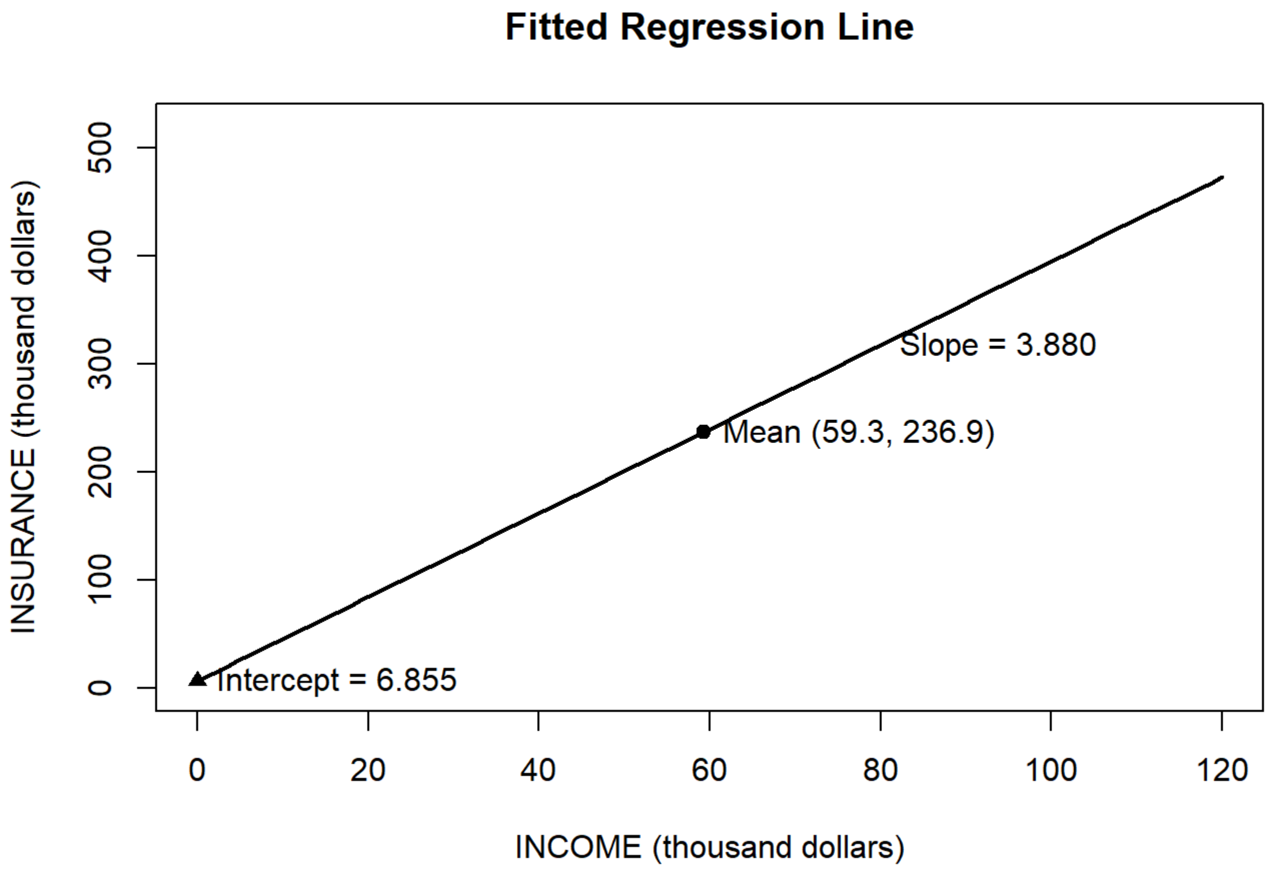
$\because$ $t > t_{0.01, 18} = 2.552$

$\therefore$ reject H_0

$\Rightarrow$ There is sufficient evidence to conclude that when income increases, the growth in life insurance holding exceeds the growth in income at 1% significance level

HW0317

3.18(a)



3.23(a)

```
collegetown <- read.csv("collegetown.csv")
model <- lm(price ~I(sqft^2), data = collegetown)
alpha2_hat <- coef(model)["I(sqft^2)"]
se_alpha2 <- summary(model)$coefficients["I(sqft^2)", "Std. Error"]

t_stat <- (40 * alpha2_hat - 13) / (40 * se_alpha2)
df <- model$df.residual
t_crit <- qt(0.95, df)
p_value <- 1 - pt(t_stat, df)
cat("t =", t_stat, "\n t_crit=", t_crit, "\n p_value=", p_value, "\n")
```

```
## t = -26.72875
##  t_crit= 1.647919
##  p_value= 1
```

在 5% 顯著水準下，計算得到的檢定統計量為 $t = -26.73$，明顯小於右尾檢定的臨界值 $t_{cr} = 1.648$。此外，p-value 為 1（大於 0.05）。因此，我們不拒絕虛無假設。由此可知，沒有足夠證據顯示在房屋面積為 2000 平方英尺時，增加 100 平方英尺所帶來的預期房價邊際效果會大於 $13,000$。

3.23(b)

```
t_stat_b <- (80 * alpha2_hat - 13) / (80 * se_alpha2)

df <- model$df.residual
t_crit_b <- qt(0.95, df)
p_value_b <- 1 - pt(t_stat_b, df)

cat("t =", t_stat_b,
    "\n t_crit=", t_crit_b,
    "\n p_value=", p_value_b, "\n")
```

```
## t = 4.189467
##  t_crit= 1.647919
##  p_value= 1.65395e-05
```

在 5% 顯著水準下，計算得到的檢定統計量為 $t = 4.819$，大於右尾檢定的臨界值 $t_{cr} = 1.648$。此外，p-value 約為 1.65×10^{-5}，遠小於 0.05。因此，我們拒絕虛無假設。 由此可知，在房屋面積為 4000 平方英尺時，增加 100 平方英尺所帶來的預期房價邊際效果顯著大於 $13{,}000$。

3.23(c)

```
collegetown <- read.csv("collegetown.csv")
model <- lm(price ~ I(sqft^2), data = collegetown)

sqft0 <- 20
new_data <- data.frame(sqft = sqft0)

pred <- predict(model, newdata = new_data, interval = "confidence", level = 0.95)

cat("fit =", pred[1,1],
    "\n lwr =", pred[1,2],
    "\n upr =", pred[1,3], "\n")
```

```
## fit = 167.3735
##  lwr = 158.0481
##  upr = 176.6988
```

根據(a)中的二次迴歸模型，當房屋面積為 2000 平方英尺時，其預期房價估計值為 167.3735（千美元）此外，95% 信賴區間為 $(158.0481,\ 176.6988)$（千美元）。

這表示我們有 95% 的信心水準認為，面積為 2000 平方英尺的房屋，其平均價格會落在 $158{,}0481$ 美元到 $176{,}6988$ 美元之間。

3.23(d)

```
houses_2000 <- subset(collegetown, sqft == 20)

n_2000 <- nrow(houses_2000)
mean_price_2000 <- mean(houses_2000$price)

cat("Number of houses =", n_2000,
    "\nSample mean price =", mean_price_2000, "\n")
```

```
## Number of houses = 3
## Sample mean price = 163.3333
```

樣本中居住面積為 2000 平方英尺（$SQFT = 20$）的房屋共有 3 間，其平均售價為 163.33（千美元），約為 $163{,}330$ 美元。

在(c)中，我們估計面積為 2000 平方英尺房屋的預期價格之 95% 信賴區間為 $(158.0481,\ 176.6988)$（千美元）。由於樣本平均售價 163.33（千美元）落在此區間內，因此可認為樣本平均售價與(c)的估計結果是相容的。

3.31(a)

```
tuna <- read.csv("tuna.csv")
summary(tuna$sal1)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1762    2485    3136    6719    8612   32820
```

```
summary(tuna$apr1)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.6100  0.6900  0.8100  0.7825  0.8625  0.9200
```

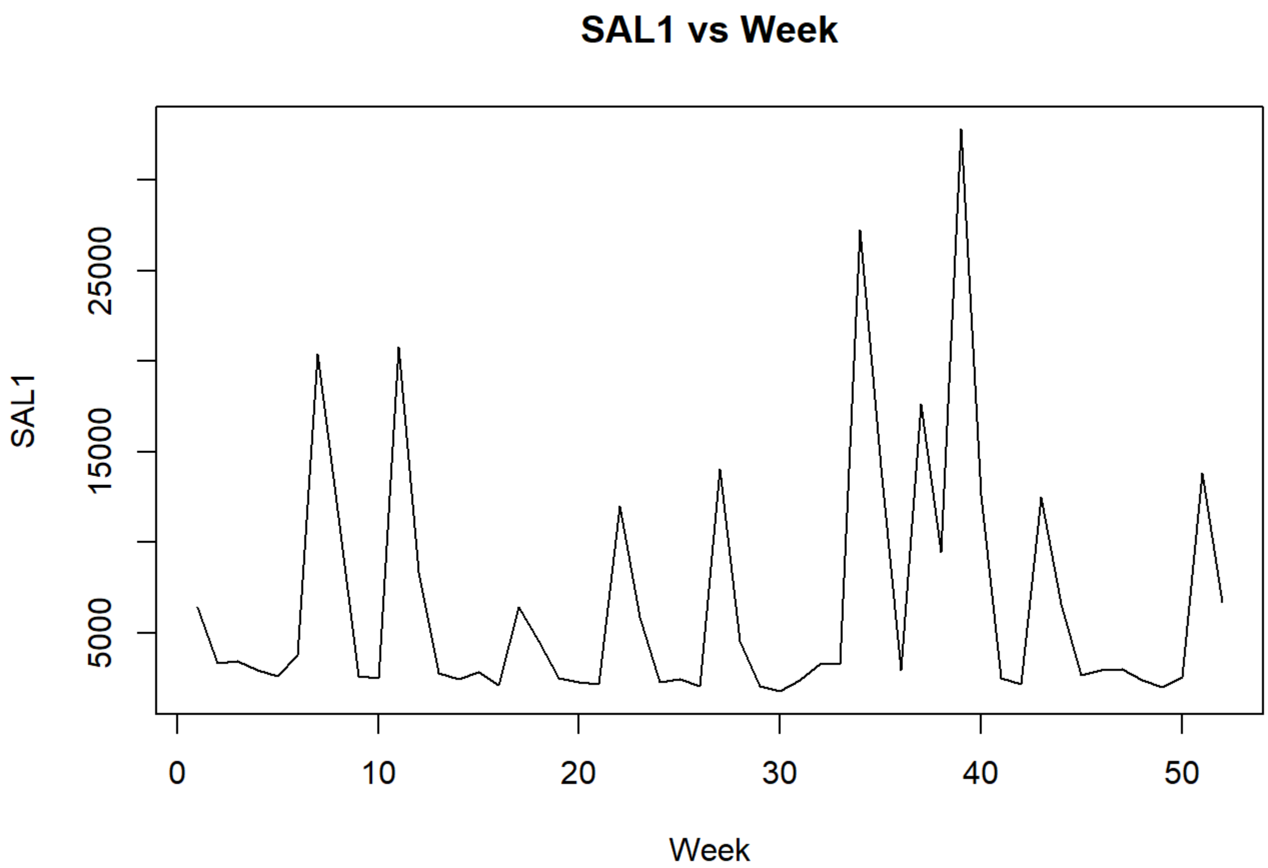
```
cat("\nstandard deviation (sal1) =", sd(tuna$sal1), "\n")
```

```
##
## standard deviation (sal1) = 6915.083
```

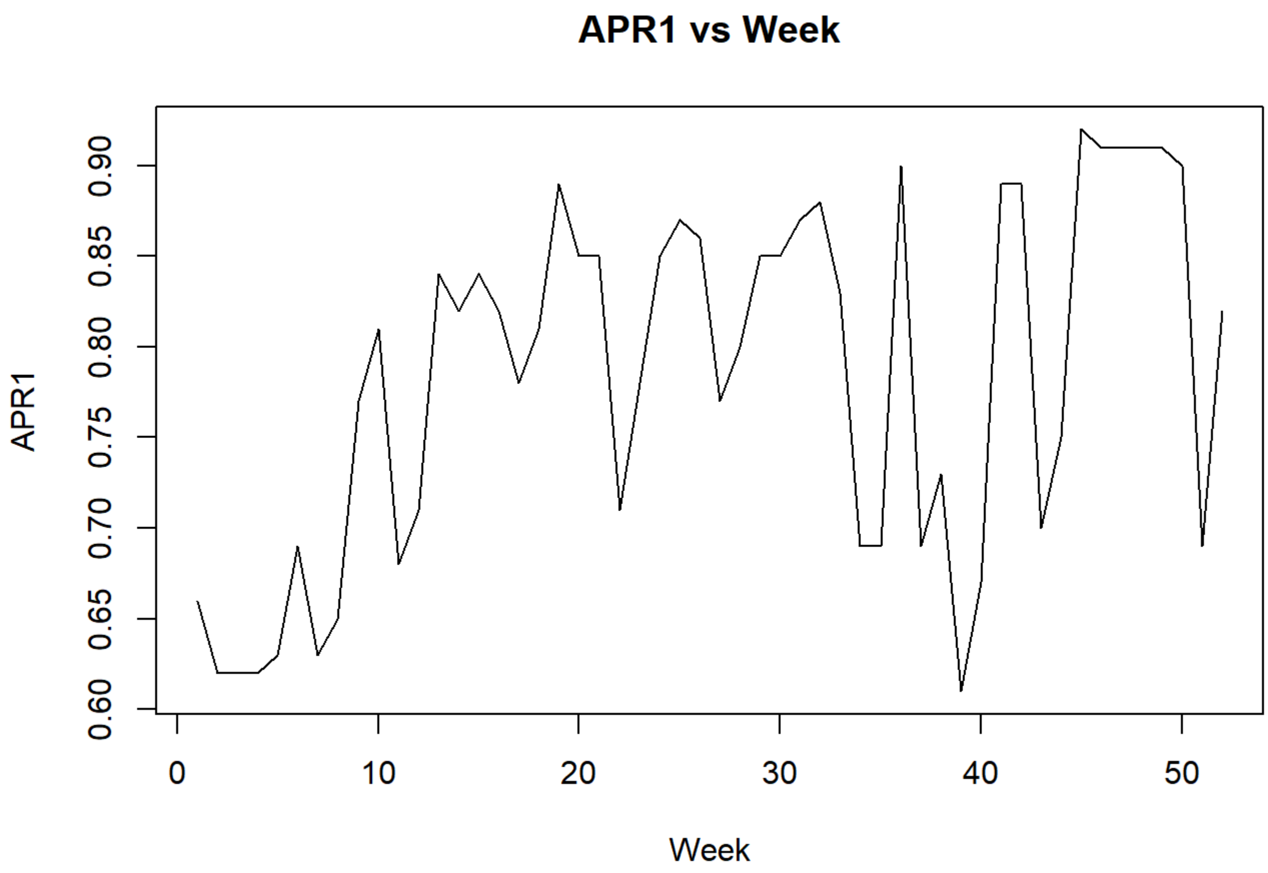
```
cat("standard deviation (apr1) =", sd(tuna$apr1), "\n")
```

```
## standard deviation (apr1) = 0.09803711
```

```
if(!"week" %in% names(tuna)) tuna$week <- 1:nrow(tuna)
plot(tuna$week, tuna$sal1,
     type = "l",
     main = "SAL1 vs Week",
     xlab = "Week",
     ylab = "SAL1")
```



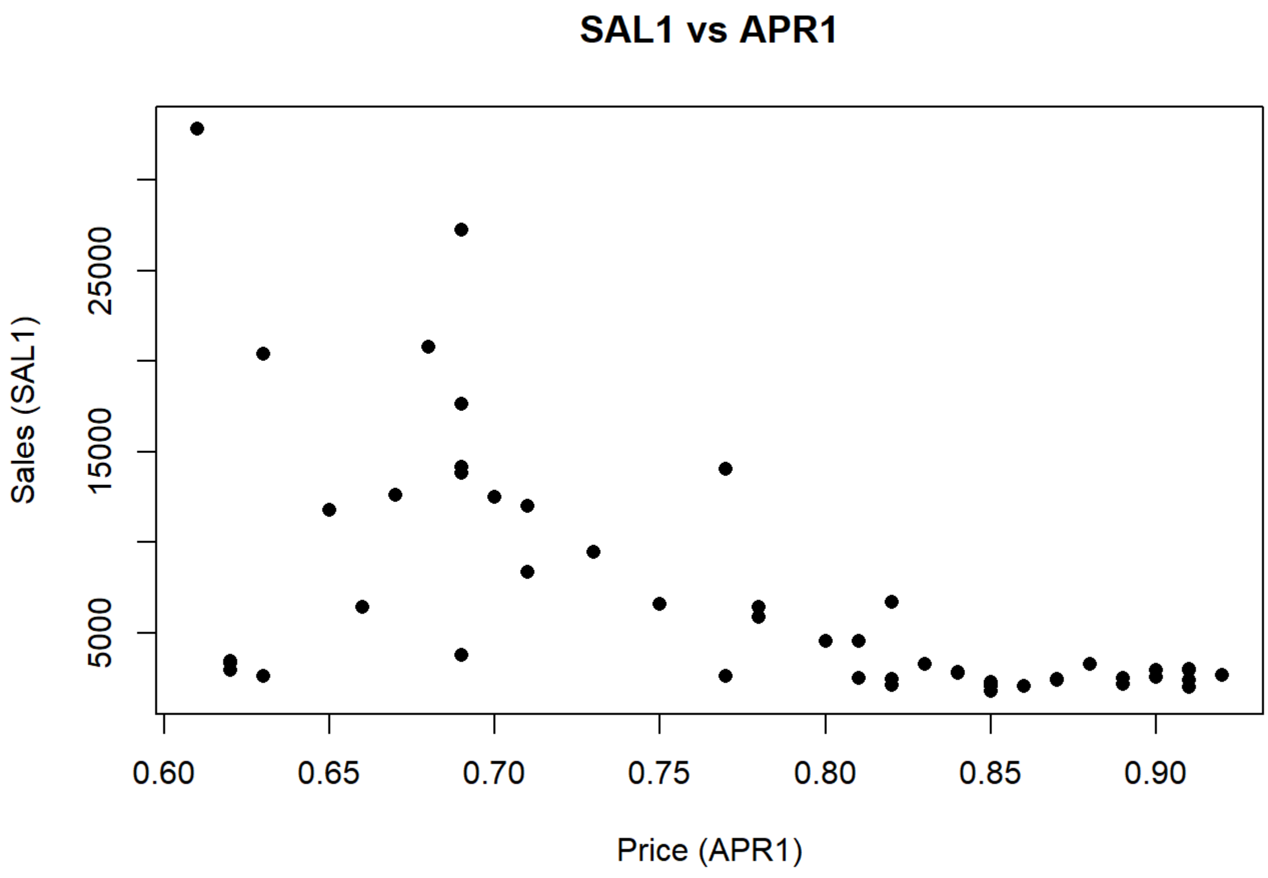

```
plot(tuna$week, tuna$apr1,
     type = "l",
     main = "APR1 vs Week",
     xlab = "Week",
     ylab = "APR1")
```



在銷售量與價格中都可以觀察到明顯的逐週變動，其中銷售量的標準差（6,915）甚至高於其樣本平均數（6,719），顯示每週銷售量具有極大的離散程度與劇烈波動，且從時間序列圖來看，整體呈現鋸齒狀且不平滑的變動，銷售量多次出現大幅向上的尖峰（最高超過 30,000 單位），代表某些週數存在短暫但劇烈的成長，而價格則呈現頻繁且明顯的下跌後快速回升的波動模式。

3.31(b)

```
plot(tuna$apr1, tuna$sal1,
     main = "SAL1 vs APR1",
     xlab = "Price (APR1)",
     ylab = "Sales (SAL1)",
     pch = 16)
```



從 SAL1 對 APR1 的散佈圖可以觀察到，兩者之間呈現明顯的反向（負）關係，即當價格（APR1）較低時，銷售量（SAL1）通常較高，而當價格上升時，銷售量則顯著下降，且整體資料點呈現由左上往右下的趨勢。此外，在較低價格區間（約 0.6 至 0.7）時，銷售量的變動幅度較大，甚至出現超過 30,000 單位的高銷售量，而在較高價格區間（約 0.8 以上）時，銷售量則普遍維持在較低水準。這樣的結果符合經濟學中的需求法則，即價格上升會降低需求量，因此價格與銷售量之間呈現負向關係是合理且符合預期的。

3.31(c)

```
tuna$price1 <- 100 * tuna$apr1

model_c <- lm(sal1 ~ price1, data = tuna)

summary(model_c)
```

```
##
## Call:
## lm(formula = sal1 ~ price1, data = tuna)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10869.5  -1588.3   -499.3   1626.2  18607.1
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 40714.22    6196.42   6.571 2.82e-08 ***
## price1       -434.45      78.58  -5.528 1.17e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5502 on 50 degrees of freedom
## Multiple R-squared:  0.3794, Adjusted R-squared:  0.367
## F-statistic: 30.56 on 1 and 50 DF,  p-value: 1.172e-06
```

```
beta2_hat <- coef(model_c)["price1"]
se_beta2 <- summary(model_c)$coefficients["price1", "Std. Error"]

df <- model_c$df.residual
t_crit <- qt(0.975, df)

lower <- beta2_hat - t_crit * se_beta2
upper <- beta2_hat + t_crit * se_beta2

cat("beta2 =", beta2_hat,
    "\n95% CI = [", lower, ",", upper, "]\n")
```

```
## beta2 = -434.4473
## 95% CI = [ -592.2897 , -276.6049 ]
```

根據迴歸模型 $SAL1 = \beta_1 + \beta_2 PRICE1 + e$ 的估計結果，當品牌一鮪魚罐頭的價格每上升 1 分（1 cent）時，其銷售量平均會減少約 434.45 單位，這就是價格每增加一分對銷售量影響的點估計。

此外，β_2 的 95% 信賴區間為 $[-592.29, \; -276.60]$。這表示我們有 95% 的信心水準認為，品牌一鮪魚罐頭的價格每增加 1 分時，其銷售量平均會減少約 276.60 到 592.29 單位。

由於點估計與整個信賴區間都為負值，顯示價格上升對銷售量具有明顯的負向影響，這與需求法則一致。

3.31(d)

```
tuna <- read.csv("tuna.csv")

tuna$price1 <- 100 * tuna$apr1

model_c <- lm(sal1 ~ price1, data = tuna)

new_data <- data.frame(price1 = 70)

pred_d <- predict(model_c, newdata = new_data,
                  interval = "confidence", level = 0.90)

cat("fit =", pred_d[1,1],
    "\n lwr =", pred_d[1,2],
    "\n upr =", pred_d[1,3], "\n")
```

```
## fit = 10302.9
##  lwr = 8624.934
##  upr = 11980.87
```

根據迴歸模型，當價格為 70 cents 時，每週預期銷售量的估計值為 $10{,}302.90$ 單位。此外，其 90% 信賴區間為 $(8{,}624.93,\ 11{,}980.87)$。

這表示我們有 90% 的信心水準認為，當價格為 70 cents 時，平均每週銷售量會落在約 8,625 到 11,981 單位之間。

3.31(e)

```
tuna$price1 <- 100 * tuna$apr1

model_c <- lm(sal1 ~ price1, data = tuna)

beta2_hat <- coef(model_c)["price1"]
se_beta2 <- summary(model_c)$coefficients["price1", "Std. Error"]

mean_price1 <- mean(tuna$price1)
mean_sal1 <- mean(tuna$sal1)

elasticity <- beta2_hat * (mean_price1 / mean_sal1)

se_elasticity <- se_beta2 * (mean_price1 / mean_sal1)

df <- model_c$df.residual
t_crit <- qt(0.975, df)

lower <- elasticity - t_crit * se_elasticity
upper <- elasticity + t_crit * se_elasticity

cat("elasticity =", elasticity,
    "\n95% CI = [", lower, ",", upper, "]\n")
```

```
## elasticity = -5.059825
## 95% CI = [ -6.898149 , -3.221501 ]
```

根據估計結果，在平均值處（at the means），品牌一銷售量對價格的彈性估計為 -5.06。其 95% 信賴區間為 $(-6.90, \ -3.22)$。

由於彈性為負值，表示價格上升會導致銷售量下降，符合需求法則。此外，彈性的絕對值大於 1，顯示需求具有高度彈性（elastic），代表消費者對價格變動相當敏感。

此結果在經濟上是合理的，因為罐頭鮪魚屬於競爭性商品，市場上存在許多替代品，當價格上升時，消費者容易轉向其他品牌或產品，因此導致銷售量大幅下降。

3.31(f)

```
mean_price1 <- mean(tuna$price1)
mean_sal1 <- mean(tuna$sal1)

elasticity_hat <- beta2_hat * (mean_price1 / mean_sal1)
se_elasticity <- se_beta2 * (mean_price1 / mean_sal1)

t_stat_f <- (elasticity_hat - (-3)) / se_elasticity
df <- model_c$df.residual
t_crit_f <- qt(0.95, df)
p_value_f <- 2 * (1 - pt(abs(t_stat_f), df))

cat("t =", t_stat_f,
    "\n t_crit =", t_crit_f,
    "\n p_value =", p_value_f, "\n")
```

```
## t = -2.250572
##  t_crit = 1.675905
##  p_value = 0.02884204
```

我們檢定在平均值處的價格彈性是否等於 -3，其虛無假設與對立假設為：

$$H_0: \ \text{elasticity} = -3$$

$$H_1: \ \text{elasticity} \neq -3$$

在 10% 顯著水準下，計算得到的檢定統計量為 $t = -2.25$，其絕對值大於臨界值 $t_{cr} = 1.676$。此外，p-value 約為 0.0288，小於 0.10。因此，我們拒絕虛無假設。

由此可知，有足夠證據顯示品牌一銷售量對價格的彈性不等於 -3。且根據前一小題的估計結果（約為 -5.06），可進一步推論其需求對價格的反應程度比 -3 更大，顯示消費者對價格變動相當敏感。